

Linear Relations

Module 5 Test

Grade 9 • MTH1W

Time: 60 minutes · Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Find the first differences for the y-values 1, 5, 9, 13. **[1]**

2. Is $y = 8x$ direct or partial variation? **[1]**

3. Is $y = 2x - 5$ direct or partial variation? **[1]**

4. Find the slope of the line through (0, 0) and (3, 12). **[1]**

5. State the y-intercept of $y = 6x + 9$. **[1]**

6. State the slope of the line $y = -3x + 2$. **[1]**

7. Find the x-intercept of $y = x - 4$. **[1]**

8. Write $y = 5x + 2$ in the standard form $Ax + By + C = 0$. **[1]**

9. A scatter plot rises steadily to the right. Is the correlation positive or negative? [1]

10. Find the first differences of 2, 8, 14, 20 and state whether the relation is linear. [2]

11. Find the slope of the line through (1, 2) and (4, 14). [2]

12. Find the x-intercept and y-intercept of $y = 2x - 8$. [2]

13. Write the equation of the line with slope 3 and y-intercept -5. [2]

14. A partial variation has initial value 10 and rate of change 4. Write its equation. [2]

15. Write $y = (3/4)x + 2$ in standard form with integer coefficients. [2]

Section B — Application

16. Find the slope of the line through (-2, 7) and (2, -1). [2]
17. Find the x-intercept and y-intercept of $2x + 5y = 20$. [2]

18. Find the equation of the line with slope 5 that passes through (2, 7). **[2]**
19. Find the first differences of 4, 9, 16, 25 and state whether the relation is linear. **[2]**
20. Write $y = (3/4)x - 2$ in standard form with integer coefficients. **[2]**
21. A direct variation passes through (6, 24). Find k and write the equation. **[2]**
22. Find the equation of the line through (1, 4) and (5, 16). **[3]**
23. A line has x-intercept 3 and y-intercept -6. Find its slope and its equation. **[3]**

24. A pool is filled at 12 L/min starting from 30 L: $V = 12t + 30$. Find V at $t = 10$ min and state the initial value. **[3]**

Section C — Thinking

25. The line through $(2, 9)$ and $(k, 21)$ has slope 4. Find k . **[4]**
26. A linear table gives $y = -5, ?, 7, 13$ for $x = 0, 1, 2, 3$. Find the missing value and the equation of the relation. **[4]**

27. A phone plan charges a \$30 monthly fee plus \$0.20 per minute. Write the cost C in terms of minutes m , find the cost for 150 minutes, and find how many minutes give a \$50 bill. (Very hard.) **[5]**
28. A line passes through $(-3, -1)$ and $(3, 11)$. Find its slope, its equation, and its x -intercept. (Very hard.) **[5]**
29. Line A is $y = 2x - 1$. Line B passes through $(1, 7)$ and $(4, 4)$. Find the equation of Line B, then find the point where the two lines intersect. (Very hard.) **[6]**

Section D — Communication

30. A student claims "every straight-line relation is a direct variation." Using an example, explain why this is false, and state the condition for a linear relation to be a direct variation. (Very hard.) **[4]**

— END OF EXAM —